Name: Vaibhav Gupta Email: vgupta1@presidio.com Date: Thursday, May 28, 2026

Manufacturing Production Monitoring System

Objective:

Develop a complete enterprise-grade application using layered architecture with PostgreSQL, ASP.NET Core Web API, and Angular frontend.

Core Modules:

- Production Planning
- Machine Monitoring
- Defect Tracking

Key Features:

- IoT simulation
- Production analytics

Suggested Architecture:

Angular Frontend → ASP.NET Core Web API → Service Layer → Repository Layer →
PostgreSQL Database

Azure Integration:

- Azure IoT simulation
- Azure Service Bus

Student Deliverables:

- Requirement Document
- ER Diagram
- API Documentation
- Angular Screens
- Authentication & Authorization
- CRUD APIs
- Search & Pagination
- Deployment Steps

Phase 2 Enhancements:

- Kubernetes

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Step-by-Step Learning Path

1. First Understand the Overall Domain (MOST IMPORTANT)

Start with these beginner-friendly guides:

Beginner MES (Manufacturing Execution System) Guides

- Rockwell Automation MES Beginner Guide
  Very beginner friendly. Explains:
  - What MES is
  - Production workflow
  - Factory operations
  - Real-time monitoring
- MachineMetrics MES Guide
  Excellent practical explanation with dashboards/screens. (machinemetrics.com)
- Tulip MES Guide
  Modern explanation of Industry 4.0 + MES concepts. (Tulip)

2. Learn Production Monitoring

This is the CORE of your project.

Best Articles

- MachineMetrics Production Monitoring
  Shows real dashboards and machine monitoring screens. (machinemetrics.com)
- Vorne Production Monitoring Guide
  Very easy explanation with examples. (Vorne)
- Evocon Production Monitoring Guide
  Explains implementation step-by-step. (Evocon)
- MaintMaster Manufacturing Monitoring
  Explains dashboards and reporting. (Maintmaster)

3. Understand OEE (Very Important)

Every manufacturing monitoring system uses OEE.

OEE = Overall Equipment Effectiveness

It measures:

- Availability
- Performance
- Quality

Best OEE Resources

- FourJaw OEE Guide (FourJaw)
- Factbird OEE Components (Factbird)
- Ubidots OEE Monitoring Guide (Ubidots)

4. Real Websites / Products Similar to Your Project

https://www.boldbi.com/dashboard-examples/manufacturing/production-monitoring-dashboard/

https://samples.boldbi.com/solutions/manufacturing/production-monitoring-dashboard

https://demo.thingsboard.io/dashboard/94116b70-7ba5-11e8-9d6f-af866d0f1760?publicId=4f11d0f0-7abf-11e8-9d6f-af866d0f1760

https://eu.thingsboard.cloud/home

https://youtu.be/GY1tVzpmfBI?si=pw7bvLCP5VoTWroT

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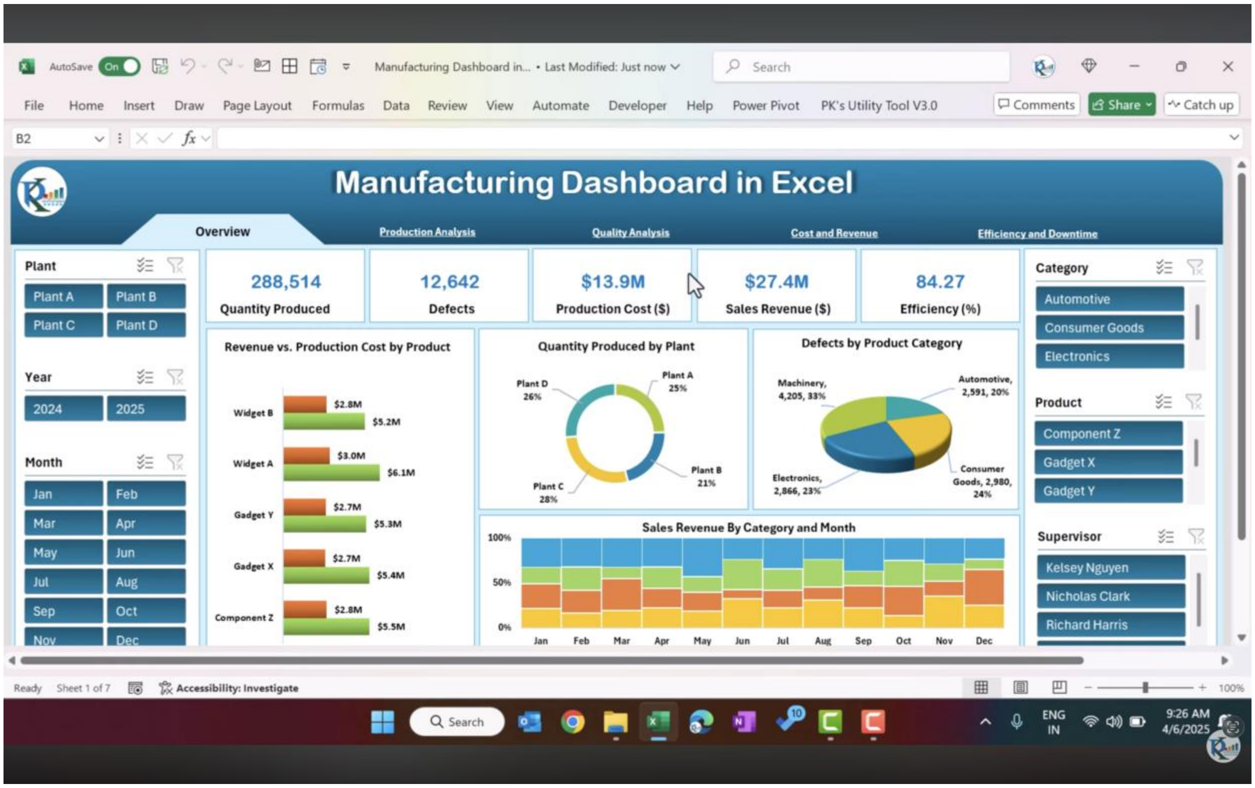

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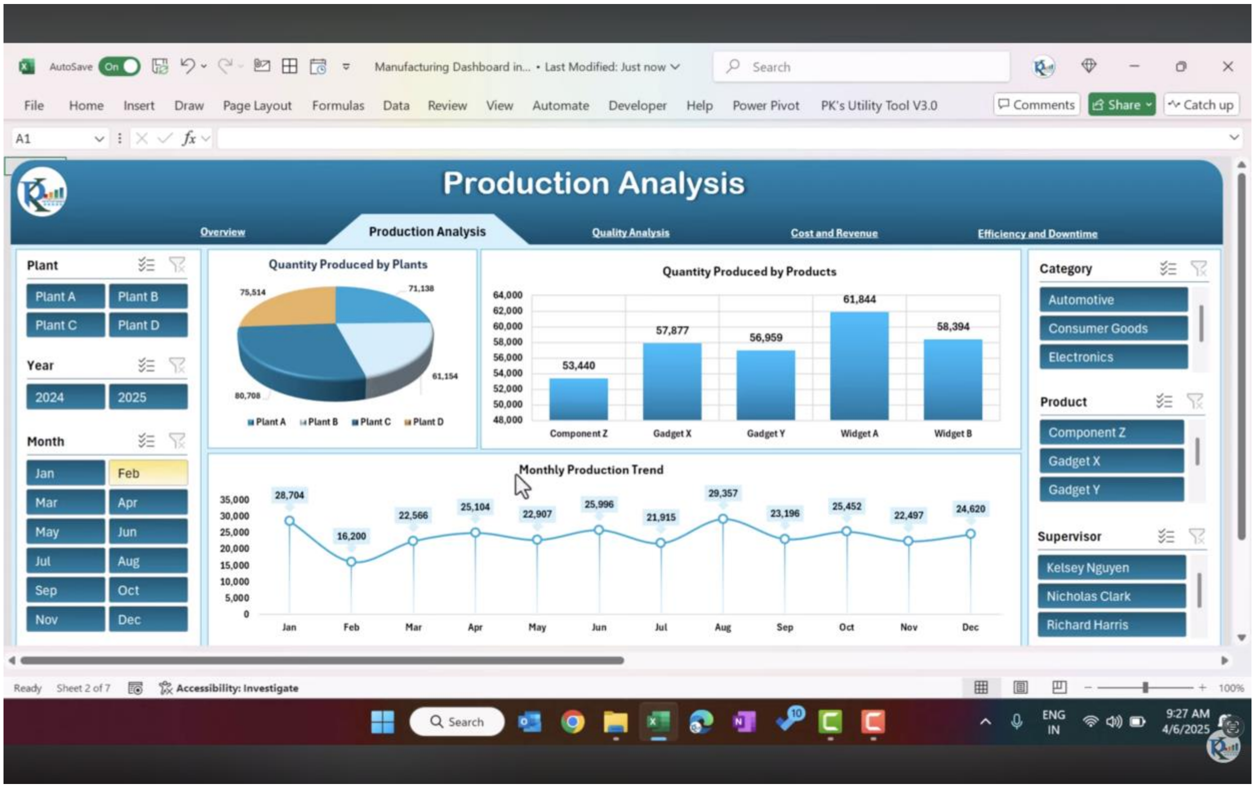

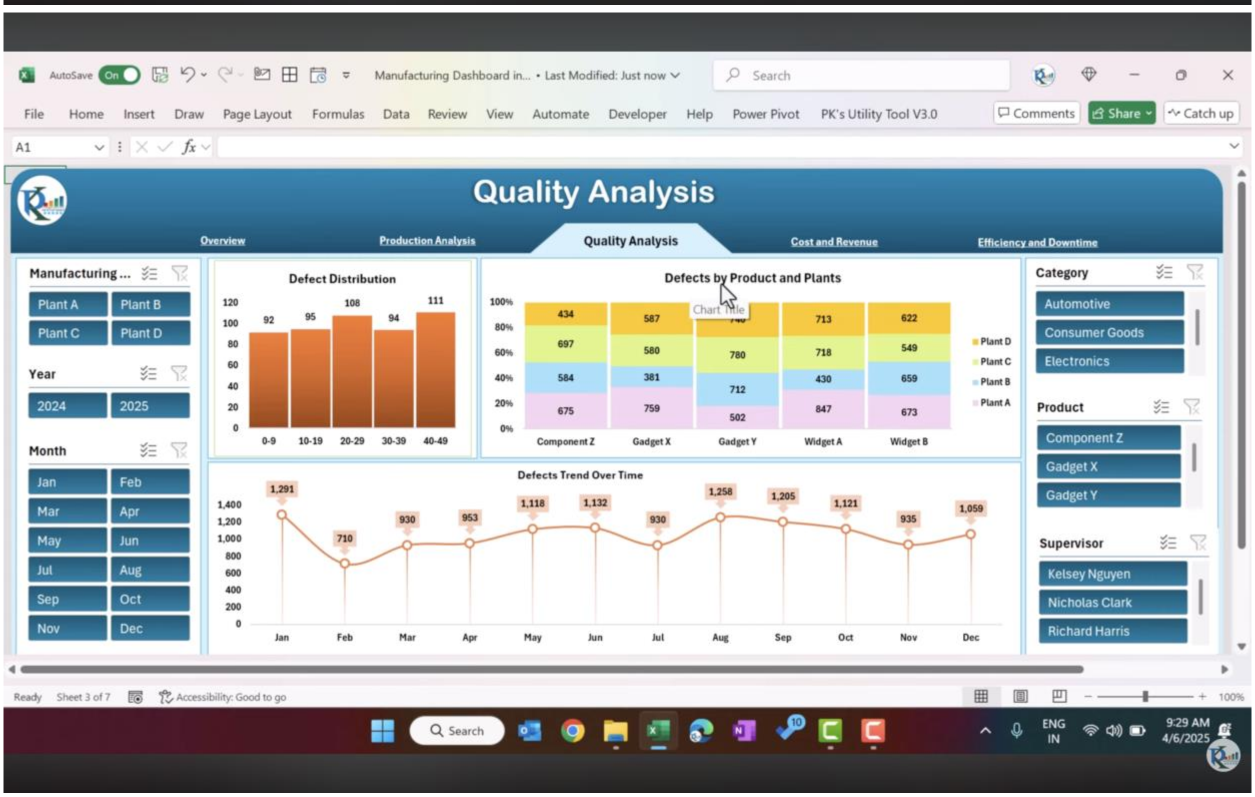

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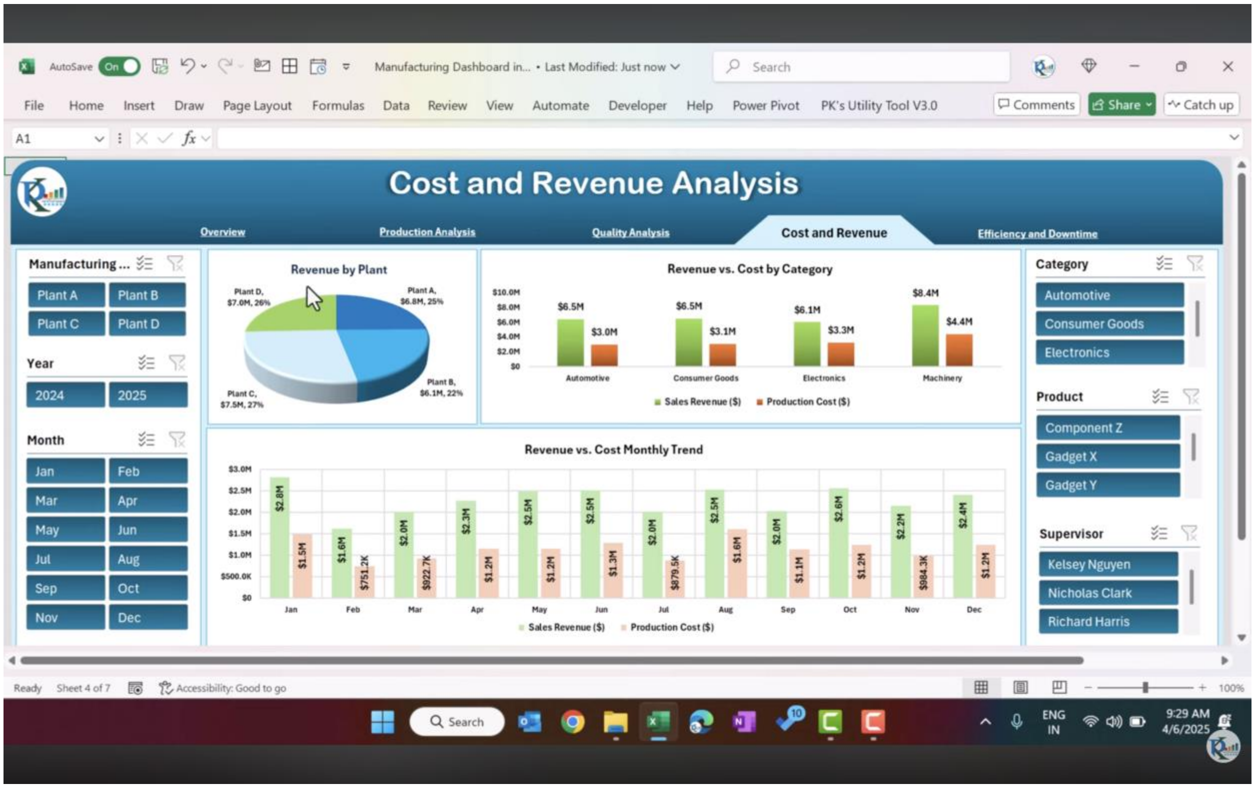

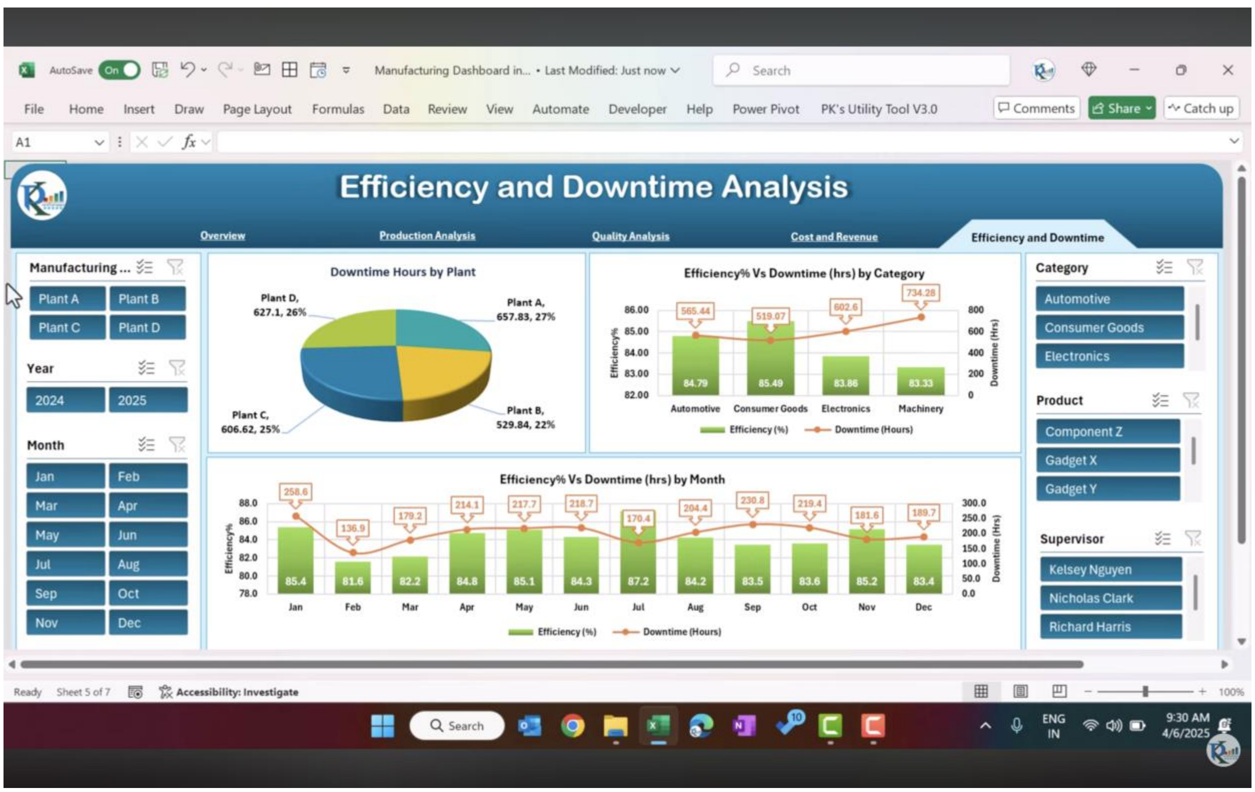

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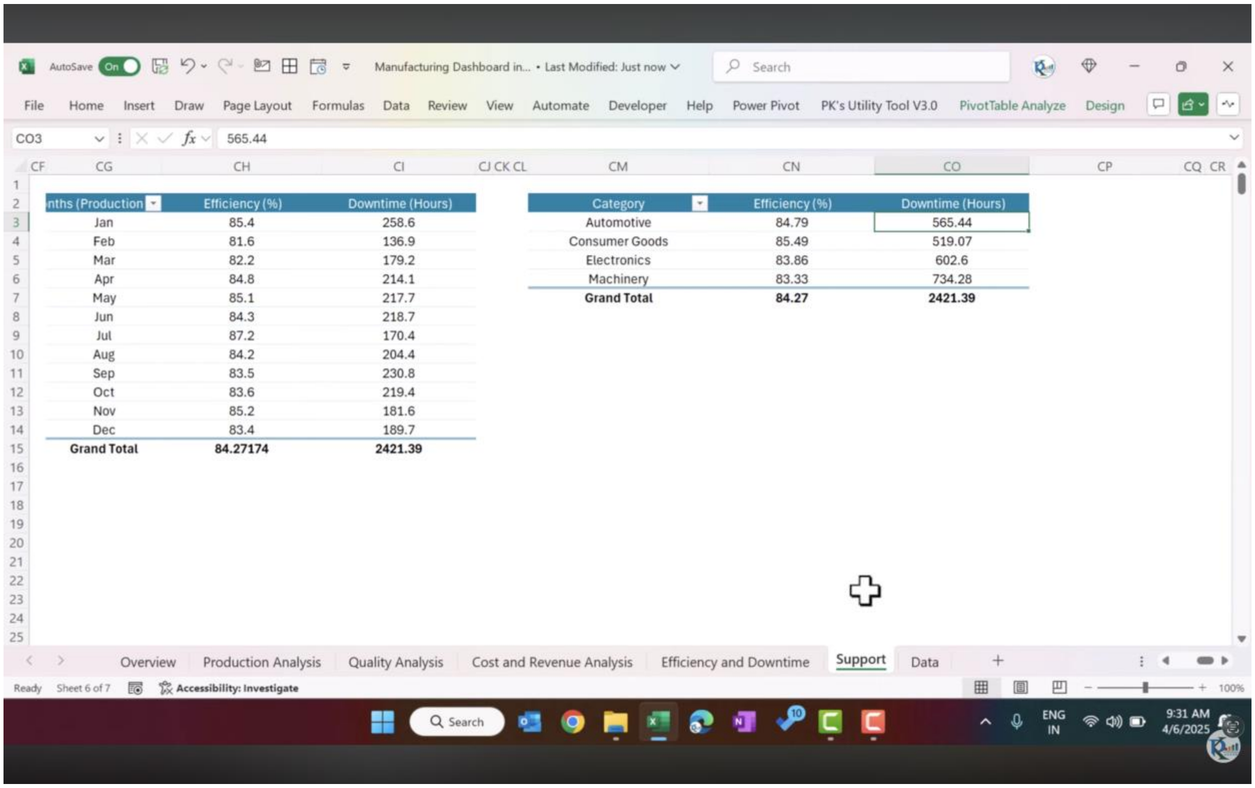
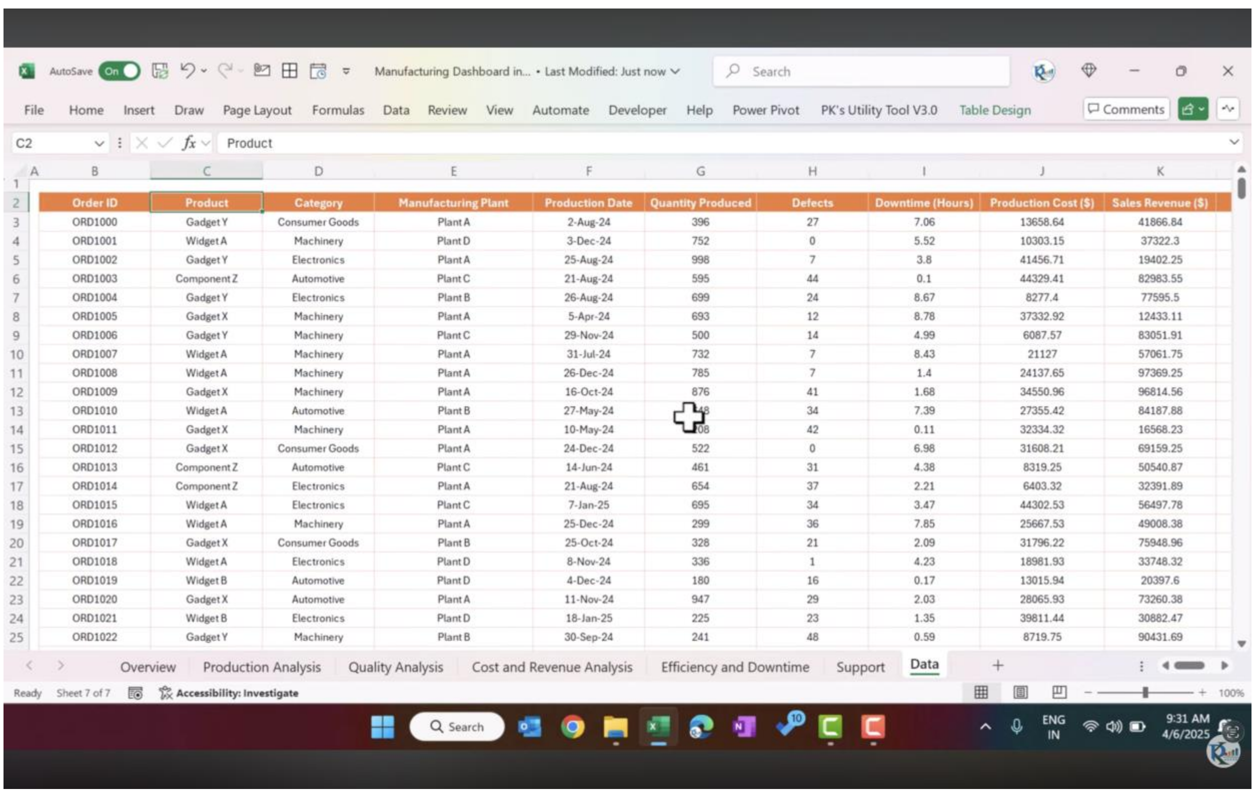


nths (Production	Efficiency (%)	Downtime (Hours)
Jan	85.4	258.6
Feb	81.6	136.9
Mar	82.2	179.2
Apr	84.8	214.1
May	85.1	217.7
Jun	84.3	218.7
Jul	87.2	170.4
Aug	84.2	204.4
Sep	83.5	230.8
Oct	83.6	219.4
Nov	85.2	181.6
Dec	83.4	189.7
Grand Total	**84.27174**	**2421.39**

Category	Efficiency (%)	Downtime (Hours)
Automotive	84.79	565.44
Consumer Goods	85.49	519.07
Electronics	83.86	602.6
Machinery	83.33	734.28
Grand Total	**84.27**	**2421.39**



Order ID	Product	Category	Manufacturing Plant	Production Date	Quantity Produced	Defects	Downtime (Hours)	Production Cost ($)	Sales Revenue ($)
ORD1000	Gadget Y	Consumer Goods	Plant A	2-Aug-24	396	27	7.06	13658.64	41866.84
ORD1001	Widget A	Machinery	Plant D	3-Dec-24	752	0	5.52	10303.15	37322.3
ORD1002	Gadget Y	Electronics	Plant A	25-Aug-24	998	7	3.8	41456.71	19402.25
ORD1003	Component Z	Automotive	Plant C	21-Aug-24	595	44	0.1	44329.41	82983.55
ORD1004	Gadget Y	Electronics	Plant B	26-Aug-24	699	24	8.67	8277.4	77595.5
ORD1005	Gadget X	Machinery	Plant A	5-Apr-24	693	12	8.78	37332.92	12433.11
ORD1006	Gadget Y	Machinery	Plant C	29-Nov-24	500	14	4.99	6087.57	83051.91
ORD1007	Widget A	Machinery	Plant A	31-Jul-24	732	7	8.43	21127	57061.75
ORD1008	Widget A	Machinery	Plant A	26-Dec-24	785	7	1.4	24137.65	97369.25
ORD1009	Gadget X	Machinery	Plant A	16-Oct-24	876	41	1.68	34550.96	96814.56
ORD1010	Widget A	Automotive	Plant B	27-May-24	[illegible]	34	7.39	27355.42	84187.88
ORD1011	Gadget X	Machinery	Plant A	10-May-24	[illegible]	42	0.11	32334.32	16568.23
ORD1012	Gadget X	Consumer Goods	Plant A	24-Dec-24	522	0	6.98	31608.21	69159.25
ORD1013	Component Z	Automotive	Plant C	14-Jun-24	461	31	4.38	8319.25	50540.87
ORD1014	Component Z	Electronics	Plant A	21-Aug-24	654	37	2.21	6403.32	32391.89
ORD1015	Widget A	Electronics	Plant C	7-Jan-25	695	34	3.47	44302.53	56497.78
ORD1016	Widget A	Machinery	Plant A	25-Dec-24	299	36	7.85	25667.53	49008.38
ORD1017	Gadget X	Consumer Goods	Plant B	25-Oct-24	328	21	2.09	31796.22	75948.96
ORD1018	Widget A	Electronics	Plant D	8-Nov-24	336	1	4.23	18981.93	33748.32
ORD1019	Widget B	Automotive	Plant D	4-Dec-24	180	16	0.17	13015.94	20397.6
ORD1020	Gadget X	Automotive	Plant A	11-Nov-24	947	29	2.03	28065.93	73260.38
ORD1021	Widget B	Electronics	Plant D	18-Jan-25	225	23	1.35	39811.44	30882.47
ORD1022	Gadget Y	Machinery	Plant B	30-Sep-24	241	48	0.59	8719.75	90431.69

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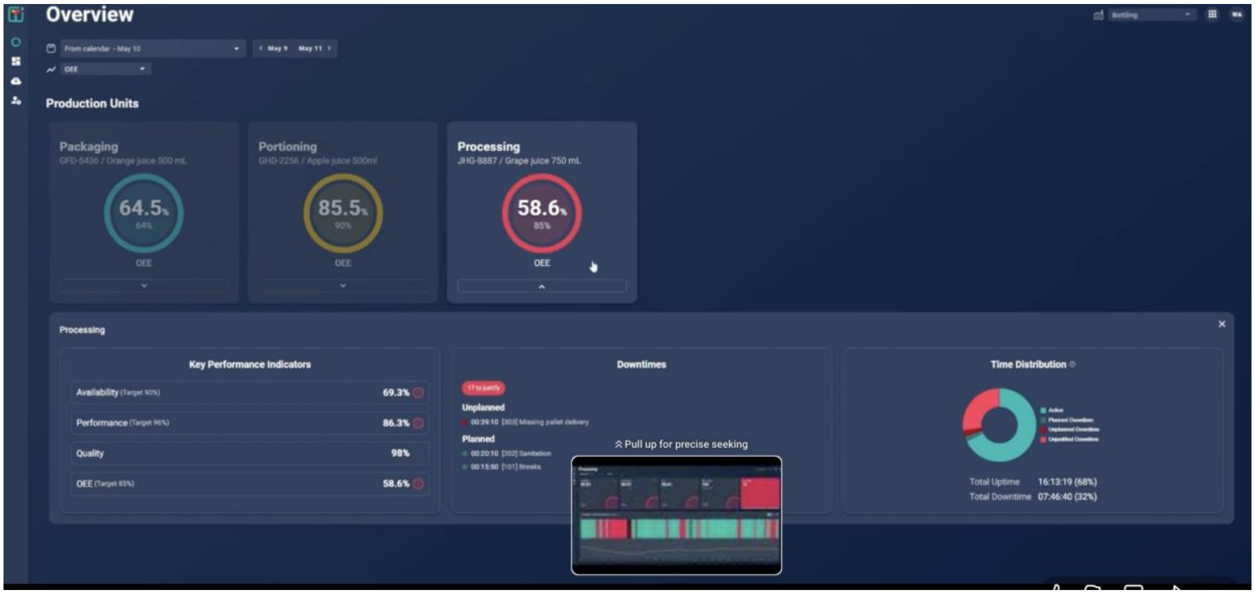
Demo of Worximity's Real-time Production Monitoring and Manufacturing Insights Software

https://youtu.be/8ChSJbWmZeM?si=MfC-jjdcAhVWywOo



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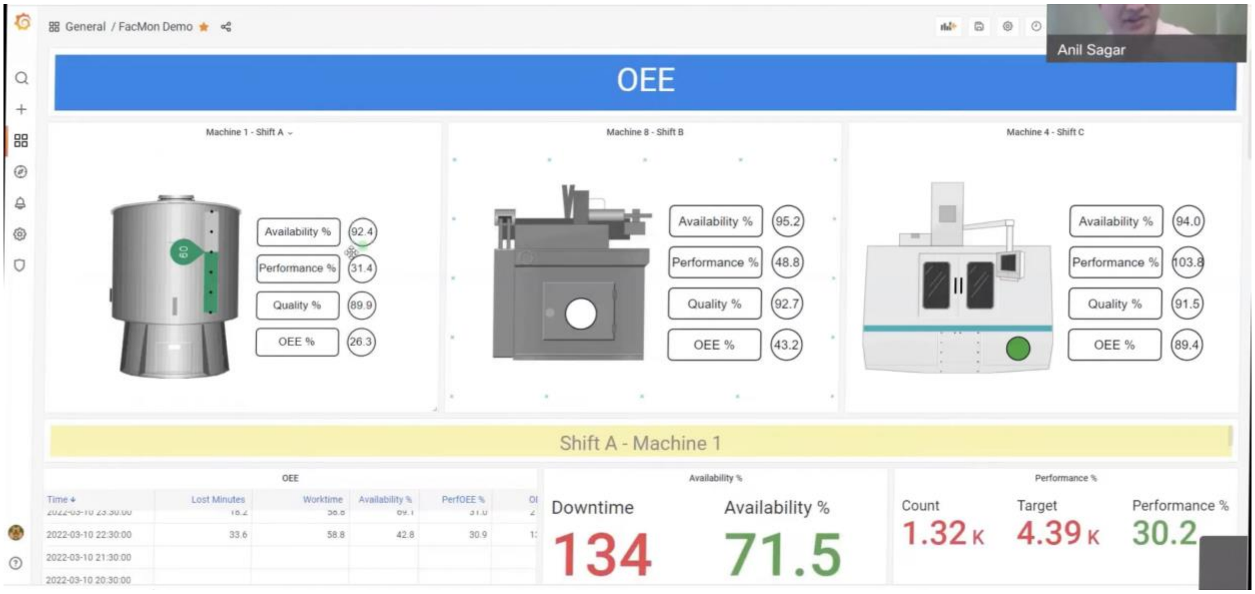

6. Understand Your Project Modules

Your project has 3 major modules.

Module 1: Production Planning

This means:

- What product to manufacture
- Quantity
- Start date
- End date
- Assigned machine
- Assigned operator

Example:

Product	Qty	Machine	Start
Bottle Caps	5000	M-101	10 AM

Module 2: Machine Monitoring

This tracks:

- Machine ON/OFF
- Temperature
- Production count
- Downtime
- Speed
- Errors

Example:

Machine	Status	Temp	Output
M-101	Running	72°C	250 units

This is where IoT simulation comes.

Module 3: Defect Tracking

Tracks:

- Damaged products
- Defect reason
- Which machine caused defect
- Defect percentage

Example:

Product	Defect Type	Count
Bottle Caps	Crack	15

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7. Understand IoT Simulation

You do NOT need real hardware.

You can simulate machine data like:


```
{
  "machineId": "M101",
  "temperature": 72,
  "status": "Running",
  "productionCount": 250
}
```


Backend generates random values every few seconds.

Then Angular dashboard updates live.

That itself counts as IoT simulation.

8. Technologies You Should Learn Before Documentation

Backend

- ASP.NET Core Web API
- Entity Framework Core
- PostgreSQL
- JWT Authentication
- SignalR (for live monitoring)

Frontend

- Angular
- Angular Material
- Charts

Azure

- Azure IoT Hub basics
- Azure Service Bus basics

DevOps

- Docker

- Kubernetes basics (Phase 2)

My Recommendation for Your Project Scope

Since this is a student project:

DO:

- Simulated machine data
- Real-time dashboard
- CRUD operations
- Analytics charts
- Authentication
- Pagination/search

DON'T initially:

- Complex industrial protocols
- Real PLC integration
- Advanced AI
- Complex Azure deployment

Start simple and gradually enhance.

The architecture itself is already enterprise-level.

Suggested Angular Screens

Authentication

- Login
- Register
- Forgot password
- Role selection

Admin Dashboard

Should contain:

KPI Cards

- Total Production
- Defect %
- Running Machines
- Downtime
- Efficiency
- Today's Orders

Charts

- Hourly production
- Defect trend
- Machine utilization
- Shift performance

Tables

- Recent defects
- Machine alerts
- Production batches

Production Planning Module

Features:

- Create production order
- Assign machine
- Assign operator
- Shift planning
- Target quantity
- Deadline tracking

Machine Monitoring Module

Features:

- Live machine status
- Temperature
- RPM
- Vibration
- Downtime
- Alarm notifications
- MQTT/IoT simulation

Defect Tracking Module

Features:

- Defect categories
- Root cause
- Defect images
- Severity levels
- Rework tracking
- Quality reports

Analytics Module

Features:

- OEE dashboard
- Production trends
- Predictive maintenance
- Defect heatmaps
- Shift comparison

Suggested Technology Stack

Frontend

- Angular 17+

- Angular Material
- ngx-charts / ApexCharts

Backend

- ASP.NET Core Web API
- JWT Authentication
- Clean Architecture

Database

- PostgreSQL

IoT

- Azure IoT Hub
- MQTT simulation

Messaging

- Azure Service Bus

Deployment

- Docker
- Kubernetes (Phase 2)